

SECTION B

Question				Marking details	Marks Available					
					Total	AO1	AO2	AO3	Maths	Prac
8.	(a)			moles of $\text{K}_2\text{Cr}_2\text{O}_7 = 0.017 \text{ mol dm}^{-3}$ (1) concentration = $\frac{0.017}{0.250} = 0.0680 \text{ mol dm}^{-3}$ (1) ecf possible from incorrect M_r	2		1 1			
	(b)	(i)		$\text{Cr}_2\text{O}_7^{2-} + 14\text{H}^+ + 6\text{e}^- \rightarrow 2\text{Cr}^{3+} + 7\text{H}_2\text{O}$	1	1				
		(ii)		$\text{Cr}_2\text{O}_7^{2-} + 14\text{H}^+ + 6\text{I}^- \rightarrow 2\text{Cr}^{3+} + 7\text{H}_2\text{O} + 3\text{I}_2$ do not accept equation with electrons included	1		1			1
		(iii)	I	starch	1	1				1
			II	Moles dichromate = $0.0680 \times \frac{25}{1000} = 1.70 \times 10^{-3} \text{ mol}$ (1) $1\text{Cr}_2\text{O}_7^{2-} \equiv 3\text{I}_2 \equiv 6\text{S}_2\text{O}_3^{2-}$ so moles thiosulfate = $6 \times 1.70 \times 10^{-3} = 0.0102 \text{ mol}$ (1) concentration = $\frac{0.0102}{\frac{24.30}{1000}} = 0.420 \text{ mol dm}^{-3}$ (1) ecf possible from incorrect stoichiometry in equation in (b)(ii)	3		1 1	1	1 1	1 1

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					Total	AO1	AO2	AO3	Maths	Prac
	(c)			orange to yellow (1) oxidation state is +6 in both $\text{Cr}_2\text{O}_7^{2-}$ and CrO_4^{2-} / doesn't change from $\text{Cr}_2\text{O}_7^{2-}$ to CrO_4^{2-} (1) neutral answer – there is no reduction or oxidation	2	1 1				1 1
	(d)	(i)		$4 \times (-1382) + 2 \times (-1128) + 3 (0) - 4 \Delta_f H^\theta = 348$ (1) $\Delta_f H^\theta = -2033 \text{ kJ mol}^{-1}$ (1) ecf possible	2		1 1		1	
		(ii)		$T = \frac{\Delta H}{\Delta S} = \frac{348000}{385}$ (1) $T = 904 \text{ K}$ (1) ecf possible	2		1 1		1	
					14	4	7	3	3	7